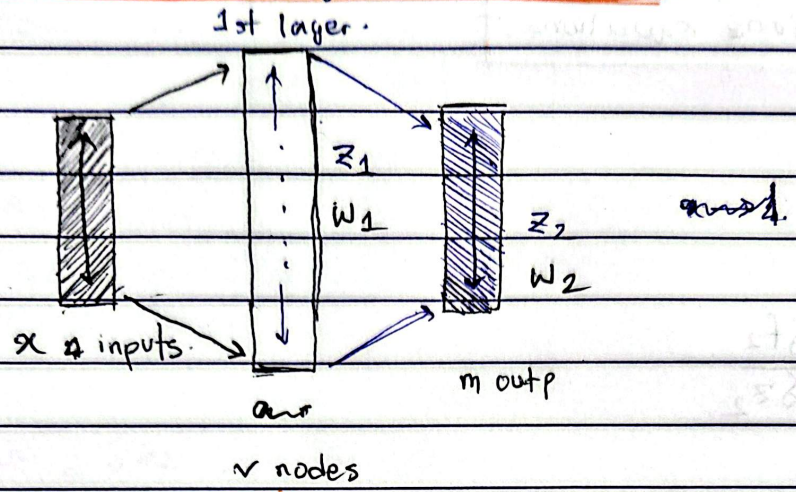
Subject Task 3

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2 layer MLP with Sigmoid activation.



Forward Pass equations

$z_1 = W_1 x + b_1$ → 1st layer

↓

$f_1 = \sigma(z_1)$

↓

$z_2 = W_2 f_1 + b_2$

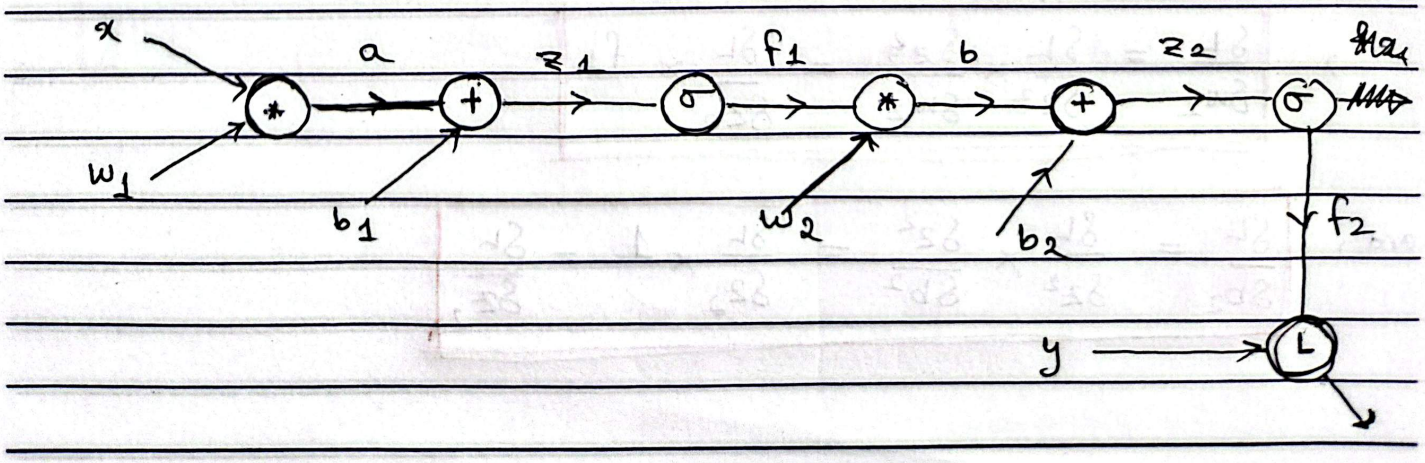
↓

$f_2 = \sigma(W z_2)$ → output predictions.

$\sigma(\cdot) = 1 / 1 + \exp(-z)$

1. Loss $= L = \frac{1}{2}(y - f_2)^2$

Computation graph :-



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Back propagation equations:-

- $\frac{\delta L}{\delta L} = 1$

$$\frac{\delta L}{\delta z_2} = \frac{\delta L}{\delta f_2} \times \frac{\delta f_2}{\delta z_2}$$

$$\frac{\delta L}{\delta f_2} = (-)\tfrac{1}{2}(2)(y - f_2) = -(y - f_2).$$

$$\frac{\delta f_2}{\delta z_2} = \sigma'(z_2) = \sigma(z_2)(1 - \sigma(z_2))$$

so, $$\frac{\delta L}{\delta z_2} = -(y - f_2)\,\sigma(z_2)(1 - \sigma(z_2))$$

$$z_2 = w_2 f_1 + b_2$$

$$\frac{\delta z_2}{\delta w_2} = f_1 \quad , \quad \frac{\delta z^2}{\delta b_2} = 1$$

so, $$\frac{\delta L}{\delta w_2} = \frac{\delta L}{\delta z^2} \times \frac{\delta z_2^2}{\delta w_2} = \frac{\delta L}{\delta z_2} \times f_1$$

and, $$\frac{\delta L}{\delta b_2} = \frac{\delta L}{\delta z^2} \times \frac{\delta z^2}{\delta b^2} = \frac{\delta L}{\delta z_2} \times 1 = \frac{\delta L}{\delta z_2}$$

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$$\frac{\delta z_2}{\delta f_1} = w_2 \quad \therefore \frac{\delta L}{\delta f_1} = \frac{\delta L}{\delta z_2} \times \frac{\delta z_2}{\delta f_1} = \frac{\delta L}{\delta z_2} w_2$$

$$f_1 = \sigma(z_1)$$

$$\frac{\delta f_1}{\delta z_1} = \sigma'(z_1) = \sigma(z_1)\,(1 - \sigma(z_1))$$

$$\frac{\delta L}{\delta f_1} = \frac{\delta L}{\delta z_2} \times \frac{\delta z_2}{\delta f_1} = \frac{\delta L}{\delta z_2} w_2$$

$$\frac{\delta L}{\delta z_1} = \frac{\delta f_1}{\delta z_1} \times \frac{\delta L}{\delta f_1} = \sigma'(z_1) \frac{\delta L}{\delta z_2} w_2$$

$$z_1 = w_1 x + b_1$$

$$\frac{\delta z_1}{\delta w_1} = x \; ; \; \frac{\delta z_1}{\delta b_1} = 1$$

$$\therefore \boxed{\frac{\delta L}{\delta w_1} = \frac{\delta L}{\delta z_1} \times \frac{\delta z_1}{\delta w_1} = \frac{\delta L}{\delta z_2} w_2 \sigma'(z_1) x}$$

$$\boxed{\frac{\delta L}{\delta b_1} = \frac{\delta L}{\delta z_1} \times \frac{\delta z_1}{\delta b_1} = \frac{\delta L}{\delta z_2} w_2 \sigma'(z_1)}$$

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negative log likelihood for a gaussian distribution.

1) we have N scalar data points:

$x_1, x_2, x_3, x_4, \ldots, x_N$

2) Assuming these datapoints were generated by a 1D normal distribution, we must estimate μ and σ.

$p(x_n \mid \mu, \sigma)$ → Probability of observing x_n given a specific μ and σ.

$$p(x_n \mid \mu, \sigma) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{\sigma^2}\right)$$

3) We want to maximize the probability of observing each x_n for some μ and σ.

$$-\ln(\text{MLE}) = -\ln\left(\prod_{n=1}^{N} p(x_n \mid \mu, \sigma)\right)$$

⇒ negative log likelihood $= -\ln(\text{MLE}) = L(\mu, \sigma)$

$$L(\mu, \sigma) = -\sum_{n=1}^{N} \ln p(x_n \mid \mu, \sigma) = -\sum_{n=1}^{N} \ln\left(\frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{\sigma^2}\right)\right)$$

$$\Rightarrow L(\mu, \sigma) = -\sum_{n=1}^{N} -\ln\sigma + \ln\left(\frac{1}{\sqrt{2\pi}}\right) - \frac{(x-\mu)^2}{\sigma^2}$$

$$\Rightarrow L(\mu, \sigma) = \sum_{n=1}^{N} \ln\sigma + (x-\mu)^2/\sigma^2 + (\text{const}).$$

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4) Partial derivatives wrt σ and μ.

$$\frac{\delta L(\mu,\sigma)}{\delta \sigma} = \sum_{n=1}^{N} \frac{1}{\sigma} - \frac{(x_n-\mu)^2}{\sigma^3} = \frac{N}{\sigma} - \frac{1}{\sigma^3}\sum (x_n-\mu)^2$$

Avg Loss derivative $= \frac{1}{\sigma} - \frac{1}{N\sigma^3}\sum (x_n-\mu)^2$

Similarly for μ)

$$\frac{\delta L}{\delta \mu} = -\frac{1}{N\sigma^2}\sum_{n=1}^{N} (x_n-\mu)$$

5) Update rules

i) $\mu^{t+1} = \mu^t - \alpha \frac{\delta L}{\delta \mu}$ → computed using $\mu^{(t)}$ and $\sigma^{(t)}$

ii) $\sigma^{t+1} = \sigma^t - \alpha \frac{\delta L}{\delta \sigma}$

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6) Deriving MLE for μ and σ^2. (finding closed form)

$$\delta L/\delta \sigma = \frac{N}{\sigma} - \frac{1}{\sigma^3}\sum (x_n - \mu)^2 = 0$$

$$+\frac{1}{\sigma^3}\sum (x_n - \mu)^2 = +\frac{N}{\sigma}$$

$$\sum (x_n - \mu)^2 = N\sigma^3/\sigma = N\sigma^2$$

$$\therefore \boxed{\sigma^2 = \frac{\sum (x_n - \mu)^2}{N}}$$

$$\frac{\delta L}{\delta \mu} = -\frac{1}{N\sigma^2}\sum (x_n - \mu) = 0$$

$$\Rightarrow \sum (x_n - \mu) = 0$$

$$\Rightarrow \sum x_n - \sum_{n=1}^{N} \mu = 0$$

$$\Rightarrow \sum x_n = N\mu$$

$$\therefore \boxed{\mu = \frac{\sum x_n}{N}}$$

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KL Divergence

$X = \{x_1, x_2, x_3, \cdots, x_n\}$

$P_\theta(x_i)$ $q_\theta(x_i)$ are 2 diff. distributions.

log likelihood :- $\log p_\theta(x_i)$, $\log q_\theta(x_i)$

similarity :- $\log P - \log Q$

$\Rightarrow \log\left(\frac{P_\theta(x_i)}{Q_\theta(x_i)}\right)$ → the log likelihood ratio

↑ the fraction is called the likelihood ratio measuring how similar 2 distributions are.

we are more interested in the average difference between P_θ and Q_θ with random values, an avg is basically termed as expected value.

$E_{P_\theta}[X] = \sum x_i P_\theta(x_i)$ ← weighted avg.

$E_{P_\theta}[h(X)] = \sum h(x_i) P_\theta(x_i)$

Expected value formulation.

for continuous RV:

$$E_{P_\theta}[h(x)] = \int_R h(x) P_\theta(x) dx$$

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Now, we can think of the log likelihood ratio as a function of random variables.

$$\text{let} \quad h(x_i) = \log\left[\frac{P_\theta(x_i)}{Q_\phi(x_i)}\right]$$

So expected value of loglikelihood ratio is:-

$$E_{P_\theta}[h(x_i)] = \sum P_\theta(x_i) \log\left(\frac{P_\theta(x_i)}{Q_\phi(x_i)}\right) \quad \Big\} \text{ for discrete RV.}$$

$$\text{for continuous RV} = \int P_\theta(x) \log\left(\frac{P_\theta(x)}{Q_\phi(x)}\right) dx.$$

This is the definition of KL Divergence

For Large numbers (According to Law of Large numbers).

$$\frac{1}{N}\sum_{i=1}^{N} h(x_i) \approx E_P[h(x)] \longrightarrow$$

in words, avg of a large number of samples is close to its expected value.

So KL Divergence can be reduced down to!

$$* \approx \frac{1}{N}\sum_{i=1}^{N} \log\left(\frac{P_\theta(x_i)}{q_\phi(x_i)}\right) \quad \{$$

true for both discrete and continuous RVs.

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$D_{KL}(P_\theta \,||\, Q_\phi)$ measures the information loss or difference when when the distribution Q is used to approximate P.

$D_{KL}(P_\theta \,||\, Q_\phi) \neq D_{KL}(Q_\phi \,||\, P_\theta)$, so they are not symmetric.

Since its assymetric, its not a distance but termed as an approximation.

Conventionally, P → reference distribution.
Q → approximation.

and so,
$D_{KL}(P \,||\, Q)$ is the forward KL divergence

Cross Entropy loss from KL divergence.

$$D_{KL}(P||Q) = \sum P \log\left(\frac{P}{Q}\right)$$

$$= \sum (P \ln P - P \ln Q)$$

$$\Rightarrow -\sum P \ln Q + \sum P \ln P$$

$$\Rightarrow (-\sum P \ln Q) - (-\sum P \ln P)$$

$$\Rightarrow \text{Cross Entropy} - \text{Entropy}.$$

So minimizing cross entropy minimizes D_{KL} (KL Divergence also decreases)

divergence decreases means prediction distribution more closely approximates the reference distribution.

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in model predictions,

Since the true label distribution P comes from the dataset and does not depend on the model weights θ, its entropy $H(P)$ is constant ; so $H(P) = 0$.

$$\therefore \quad D_{KL}(P \,||\, Q_\theta) = H(P, Q_\theta) - H(P)$$

we aim to minimize D_{KL}

$$\text{so,} \quad \arg\min_\theta D_{KL}(P \,||\, Q_\theta) = \arg\min_\theta \left(H(P, Q_\theta) - 0\right)$$

thus, minimizing cross entropy minimizes D_{KL} divergence.

Zero Avoiding Property → KL Divergence heavily penalizes the approximation if it assigns zero probability to an event that the true distribution P considers possible.

Assymetric, as penalization applies to the forward KL Divergence only.